

Design and Implementation of a 32-bit Single-Cycle RISC-V Processor

Muhammad Husnain Jatoi

Department of Computer Engineering
Ghulam Ishaq Khan Institute of
Engineering Sciences and Technology
Reg number: 2024405

Irtaza Shahbaz

Department of Computer Engineering
Ghulam Ishaq Khan Institute of
Engineering Sciences and Technology
Reg number: 2024245

Muhammad Musa

Department of Computer Engineering
Ghulam Ishaq Khan Institute of
Engineering Sciences and Technology
Reg number: 2024428

Abstract—This report details the architectural design and functional implementation of a 32-bit single-cycle RISC-V microprocessor. Utilizing the standard RV32I base integer instruction set, the processor is modeled using Verilog Hardware Description Language (HDL). The datapath is partitioned into distinct modular components, including the instruction memory, register file, immediate generator, arithmetic logic unit (ALU), data memory, and control unit. Each module's theoretical foundation, internal schematic logic, and hardware realization are thoroughly analyzed. The single-cycle architecture ensures that each instruction completes within one clock cycle, providing a robust baseline for understanding microprocessor organization, control flow orchestration, and memory hierarchy interfacing.

Index Terms—RISC-V, Single-Cycle Processor, Verilog, Computer Architecture, Datapath, Control Unit, ALU.

I. INTRODUCTION

The Reduced Instruction Set Computer (RISC) paradigm revolutionized modern computer architecture by emphasizing a simplified set of instructions that execute rapidly and deterministically. RISC-V is an open-standard Instruction Set Architecture (ISA) that has gained immense traction in both academia and industry due to its modularity, extensibility, and lack of licensing fees. The RV32I base integer instruction set, utilized in this project, defines a 32-bit architecture with 32 integer registers, providing a comprehensive foundation for executing high-level software constructs.

This project implements a single-cycle RV32I processor. In a single-cycle implementation, the entire datapath—from fetching an instruction to writing back the computed result—is traversed in a single clock cycle. While this limits the maximum theoretical clock frequency due to the lengthy critical path (typically routed through the instruction memory, register file, ALU, and data memory sequentially), it offers a straightforward and deterministic model that is mathematically rigorous and ideal for foundational architectural exploration.

The implementation is broken down into modular Verilog components, directly reflecting the logical blocks of a standard microprocessor datapath. These components include the top-level routing logic, program counter, instruction fetch mechanism, instruction decode logic, immediate value generation, arithmetic operations, and data memory access mechanisms.

II. PROJECT DESCRIPTION AND ARCHITECTURE

The core objective of this project is to construct a fully functional soft-core processor capable of executing basic machine code compiled for the RISC-V target architecture. The system strictly follows the Harvard architecture principles, maintaining distinct and separate memories for instructions and data. This separation prevents structural hazards and allows the simultaneous fetching of an instruction word and accessing of a memory operand.

The design relies on a central `top` module that instantiates and wires together the entirety of the datapath. The datapath itself represents the physical manifestation of the standard instruction execution cycle: Fetch, Decode, Execute, Memory Access, and Writeback. In this single-cycle design, these five stages are compressed into purely combinational logical paths, bounded only by the sequential, synchronous elements: the Program Counter (PC) register, the Register File's write port, and the Data Memory's write port.

By orchestrating the data flow via multiplexers governed by a centralized Control Unit, the processor can dynamically reconfigure its combinatorial pathways on-the-fly to execute drastically different instructions (e.g., swapping from an addition to a memory load) in subsequent clock cycles.

III. PROJECT REQUIREMENTS AND ISA COMPLIANCE

The processor is fundamentally required to support the core instructions of the RV32I subset. This involves handling multiple distinct instruction formats encoded within the 32-bit instruction word:

- **R-Type:** Register-to-Register operations. These include arithmetic and logical functions such as ADD, SUB, SLL (Shift Left Logical), and XOR.
- **I-Type:** Register-Immediate operations, and load operations. Examples include ADDI, LW (Load Word), and SLTI (Set Less Than Immediate).
- **S-Type:** Store operations used to write to data memory, such as SW (Store Word), SH (Store Halfword), and SB (Store Byte).
- **B-Type:** Conditional branching instructions that alter control flow based on comparisons, such as BEQ (Branch if Equal) and BLT (Branch if Less Than).

- **J-Type & U-Type:** Unconditional jumps (JAL, JALR) and upper immediate operations (LUI, AUIPC) for large immediate manipulations and absolute control flow changes.

Compliance with these formats requires meticulous bit-slicing within the decode stage and a highly robust control unit capable of asserting the correct multiplexer selection lines and write-enable signals simultaneously to avoid data corruption.

IV. HARDWARE COMPONENTS SPECIFICATION

This section provides a highly detailed analysis of the individual Verilog modules comprising the processor, outlining their operational principles and coding methodologies.

A. Program Counter (PC) and Fetch Logic

The Program Counter (`program_counter`) is an archetypal 32-bit register that tracks the memory address of the currently executing instruction. It acts as the primary synchronous element initiating the instruction cycle. It resets to an initial state of zero upon receiving an active-high `rst` signal and updates to the evaluated `pc_next` on the positive edge of the system clock.

The `pc_next` value is determined dynamically by a multiplexer governed by the `pc_src` signal originating from the Control Unit. If no branch or jump is executed, the PC defaults to sequential execution, incrementing by 4 bytes (`pc_plus4 = pc + 32'd4`). In the event of a taken branch or a JAL (Jump and Link) instruction, the immediate value generated by the immediate generator is added to the current PC. For JALR (Jump and Link Register) instructions, the target address is calculated independently as the sum of a register value and an immediate. A critical principle enforced here is aligning the address by masking the least significant bit:

```
assign jalr_target = (reg_data1 + imm_out) & ~32'd1;
```

This hardware-level masking ensures the processor does not attempt to execute misaligned instructions.

B. Instruction Memory (`inst_mem`)

The Instruction Memory represents the Read-Only Memory (ROM) from the perspective of the operational datapath. It is implemented as an array of 256 32-bit words, providing 2KB of addressable storage space. Since the RISC-V architecture dictates byte-addressable memory but mandates that instructions are strictly 32-bit aligned, the memory module natively accesses words using `addr[31:2]`, effectively shifting out the lower two bits to perform word-aligned physical reads:

```
assign inst = memory[addr[31:2]];
```

The memory is populated at system initialization via the `$readmemh` system task, which loads hexadecimal machine code from an external `instructions.mem` file.

C. Instruction Decode & Register File (`reg_file`)

Once a raw 32-bit instruction is fetched from memory, it is structurally decomposed into its constituent fields via simple combinational wire assignments: `opcode` (bits 6:0), `rd` (11:7), `funct3` (14:12), `rs1` (19:15), `rs2` (24:20), and `funct7` (31:25).

The Register File is a high-speed synchronous memory structure containing 32 independent 32-bit registers. The RISC-V specification explicitly mandates that register `x0` is hardwired to a constant zero. The Verilog implementation elegantly ensures this by utilizing a ternary operator during the read phase:

```
assign read_data1 = (rs1==5'd0) ? 32'd0 : reg_file[rs1];
assign read_data2 = (rs2==5'd0) ? 32'd0 : reg_file[rs2];
```

Writes to the register file are synchronous (clock-dependent) and strictly governed by the `reg_write_en` control signal. The write address is specified by the `rd` field (`rsW` in the module port list). Synchronous writing acts as a temporal barrier, preventing race conditions and ensuring that volatile results generated by the ALU or data memory late in the clock cycle are reliably latched exactly at the beginning of the subsequent clock cycle.

D. Immediate Generation (`imm_gen`)

Immediate generation is arguably one of the most critical and structurally complex static routing tasks in the RV32I datapath. Because the ISA intentionally places immediate bits in varying locations across different instruction types to preserve the positional consistency of register indices (`rs1`, `rs2`, `rd`), the `imm_gen` module utilizes heavy concatenation and sign-extension.

The module inspects the base `opcode` to determine the specific instruction format. For an I-Type instruction, the 12-bit immediate is simply found in bits [31:20]. The module creates a 32-bit output by replicating the sign bit (bit 31) twenty times and concatenating it with the original 12 bits.

For highly scrambled formats like the B-Type (Branch), the bits are non-contiguous. The sign bit remains at 31, but the rest are distributed. The Verilog reassembles this effectively:

```
imm_out = {{19{instruction[31]}}, instruction[31],
           instruction[7], instruction[30:25],
           instruction[11:8], 1'b0};
```

It is crucial to note the explicit appending of `1'b0` at the least significant position. This shifts the generated value by one bit (multiplying by 2) because branch offsets are always measured in 2-byte increments.

E. Arithmetic Logic Unit (`alu_logic`)

The ALU acts as the computational core of the microprocessor. It receives two 32-bit operands: `op1` and `op2`. The source for `op1` is typically `reg_data1`, but for instructions like AUIPC, the control signal `a_sel` dynamically routes the PC value into `op1`. The source for `op2` is determined by

the `alu_src` multiplexer, choosing between `reg_data2` (for standard R-Type arithmetic) and the generated immediate value `imm_out` (for I-Type logic and memory addresses).

The module utilizes a 4-bit `alu_op` control signal to differentiate between multiple distinct operations:

- **4'b0000:** ADD / ADDI / AUIPC (Addition and Address Calculation)
- **4'b0001:** SUB (Subtraction)
- **4'b0010:** SLL (Shift Left Logical)
- **4'b0011:** SLT (Set Less Than, Signed Evaluation)
- **4'b0100:** SLTU (Set Less Than, Unsigned Evaluation)
- **4'b0101:** XOR / XORI (Bitwise Exclusive OR)
- **4'b0110:** SRL (Shift Right Logical)
- **4'b0111:** SRA (Shift Right Arithmetic)
- **4'b1000:** OR / ORI (Bitwise OR)
- **4'b1001:** AND / ANDI (Bitwise AND)

A fundamental principle applied here is the distinction between logical and arithmetic operations. Signed operations utilize the `$signed()` Verilog system function to ensure proper two's complement evaluation, while arithmetic right shifts use the `>>>` operator to preserve the sign bit, shifting in copies of the most significant bit rather than zeroes.

F. Branch Comparator (*branch_comp*)

To radically decouple branch condition evaluation from the main ALU computation path, a dedicated branch comparator operates in parallel. It evaluates the boolean expression `a == b` to assert the `BrEq` flag, and evaluates `a < b` to assert `BrLt`.

The `BrLt` computation respects the `BrUn` (Branch Unsigned) control signal driven by the Control Unit, dynamically switching between logical (unsigned) and arithmetic (signed) comparisons:

```
assign BrLt = BrUn ? (a < b) :
               ($signed(a) < $signed(b));
```

This hardware parallelization eliminates a severe critical path latency bottleneck that would otherwise manifest if the main ALU were tasked with sequentially calculating the target address and then evaluating the comparative condition.

G. Data Memory (*data_mem*)

The Data Memory simulates 1KB of operational RAM structured as 256 32-bit words. It handles both combinational Load (read) and synchronous Store (write) operations. A sophisticated hallmark of the RISC-V architecture is native support for sub-word (byte and half-word) memory access.

The module decodes the `funct3` field to distinguish access granularities. For a signed Byte Load (`lb`), it utilizes the lower two bits of the address (`addr[1:0]`) to select the specific 8-bit segment from the 32-bit word line. It then sign-extends this byte to a full 32 bits. Conversely, for an Unsigned Byte Load (`lbu`), the upper 24 bits are purely padded with zeroes. Half-word accesses (`lh`, `lhu`) utilize `addr[1]` to select the upper or lower 16 bits.

Store operations (`sb`, `sh`, `sw`) use precise synchronous, selective writing logic. Instead of indiscriminately overwriting the entire 32-bit word in memory, a Store Byte (`sb`) instruction uses a localized case statement to write exclusively to the targeted byte sector, preserving the remaining 24 bits of data intact.

H. Control Unit (*control_unit*)

The Control Unit is the central nervous system of the processor, generating all multiplexer selection lines, write enables, and computational operation codes. It achieves this by cascading a main decoder (evaluating the primary opcode) and an ALU subsidiary decoder (evaluating `funct3` and `funct7`).

By default, the CU initializes all control signals to zero. This defensive programming paradigm prevents the unintended inference of memory latches during Verilog synthesis and outright prevents rogue memory overwrites. When a standard R-Type opcode (`7'b0110011`) is detected, `alu_src` remains zero (selecting the second register), `reg_write_en` is asserted, and the `alu_op` is finely determined by evaluating a concatenation of `funct7` and `funct3`.

For Branch instructions (`7'b1100011`), `alu_src` is bypassed. The `branch_unsigned` signal is dynamically asserted if `funct3` specifies `BLTU` or `BGEU`. A nested evaluation block evaluates `funct3` against the external combinatorial flags `BrEq` and `BrLt` to conditionally update `pc_src` to `2'b01`, triggering a programmatic jump.

The Writeback multiplexer is intelligently controlled by the `mem_to_reg` flag (to select data memory output for Load instructions) and `pc_src` (to route the `pc_plus4` address directly into the register file for `JAL` and `JALR` linking instructions).

V. BLOCK DIAGRAM AND SIGNAL ROUTING TOPOLOGIES

The physical layout and routing paths of the processor are essential to understanding its single-cycle execution.

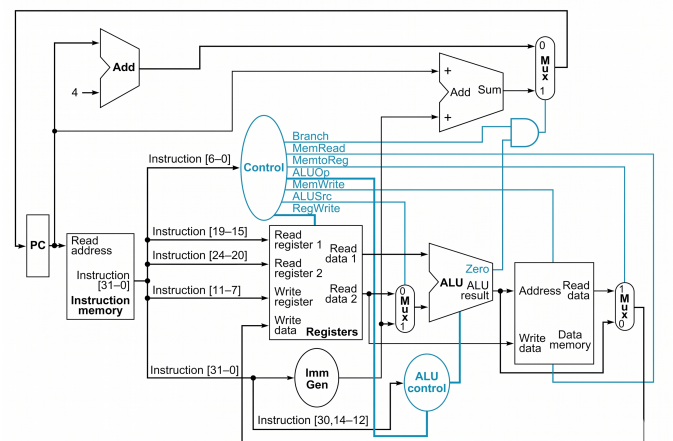


Fig. 1. Hardware block diagram illustrating the full datapath and control unit routing for the single-cycle processor.

As illustrated in Fig. 1, the structural topology is clearly defined by the instantiation hierarchy within the `top` module. The PC serves as the architectural spine. Its 32-bit output vector branches into the Instruction Memory and an independent adder computing PC+4. The fetched 32-bit instruction immediately fans out: the opcode flows to the Control Unit, the register addresses direct to the Register File, and the immediate bits route to the Immediate Generator.

The Register File outputs merge harmoniously with the immediate values at the ALU inputs via multiplexers. The ALU output routes simultaneously in three directions: to the Data Memory address port, to the Next PC logic (acting as the base address for JALR), and to the Writeback multiplexer. The processing loop is ultimately closed when the Writeback multiplexer funnels the final computed mathematical or memory value back into the Register File’s write data port.

VI. CONCEPTS AND PRINCIPLES

A. Single-Cycle Datapath Limitations and Timing Constraints

In this un-pipelined architectural design, the global clock period must be sufficiently lengthy to accommodate the absolute worst-case propagation delay. The critical path typically manifests during a Load instruction (`lw`). The combinatorial chain spans from the Clock-to-Q delay of the PC register, propagating through the Instruction Memory Read, into the Register File Read, through the ALU for address calculation, cascading into the Data Memory Read, and finally terminating at the Register File Setup Time threshold. All these consecutive combinatorial delays must mathematically sum to less than the designated clock period. While instruction throughput (IPC) is theoretically optimal at 1.0, the drastically lower maximum allowable frequency makes this design purely foundational, paving the way for advanced 5-stage pipeline implementations.

B. Data Hazards and Control Flow Resolution

Unlike advanced pipelined or superscalar architectures, single-cycle processors inherently do not suffer from data hazards (such as Read-After-Write or Write-After-Write). This is because the entire architectural state (both internal registers and external memory) is updated simultaneously exactly at the clock edge, completely finalizing an instruction’s execution state before the next instruction is ever fetched. However, programmatic control flow must be resolved purely combinatorially. The Next PC selection logic must idly wait for the Branch Comparator to fully resolve its boolean output. If a branch is successfully evaluated as taken, the multiplexer routes the calculated target address back to the PC input precisely in time for the rising clock edge to latch the new trajectory.

VII. SIMULATION AND VERIFICATION

To validate the functional correctness of the implemented single-cycle processor, a comprehensive testbench (`top_tb.v`) was developed and simulated. The testbench initializes the memory modules, supplies a steady 10ns periodic

clock, and applies an initial reset sequence to stabilize the Program Counter before commencing instruction execution.

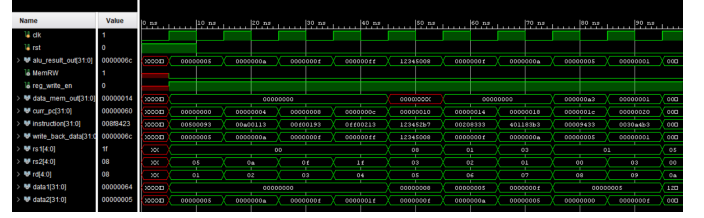


Fig. 2. Complete simulation waveform showcasing the execution of various RISC-V instructions over time, including stable clock edges and memory interactions.

As observed in the simulation waveforms (Fig. 2 and Fig. ??), the control signals correctly respond to the fetched opcodes on a per-cycle basis. For instance, during a memory store instruction, the `MemRW` signal is visibly asserted high, whereas during R-type or I-type arithmetic operations, the `reg_write_en` signal transitions high. The ALU accurately computes the expected results (visible in the `alu_result_out` vector), which are then correctly propagated to the data memory or routed back to the register file. This dynamic behavioral analysis confirms the robust integrity of the integrated datapath and control unit.

VIII. CONCLUSION

This report comprehensively presented the structural, theoretical, and functional design of a single-cycle RISC-V 32-bit processor. The HDL implementation demonstrates strict, mathematically proven adherence to the standard RV32I specification, successfully integrating complex subsystems such as sub-word bit-aligned memory access, highly combinatorial immediate sign-extension, and dynamic parallel branch evaluation. The modular Verilog code constructed herein provides an excellent pedagogical framework for analyzing digital logic at the architectural level, seamlessly translating abstract ISA specifications into tangible combinatorial and sequential digital logic constructs.

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